

ABEERA IMTIAZ

AI/ML ENGINEER |

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SUMMARY

Full-Stack Data Scientist transforming raw data into actionable insights. Skilled in ETL, ML deployment, and building AI solutions that are both scalable and visually impactful.

TECHNICAL SKILLS

- **Machine Learning:** Supervised/Unsupervised, XGBoost, Random Forest, CNN, U-Net Segmentation.
- **MLOps & Deployment:** MLflow, Docker, GitHub Actions (CI/CD), FastAPI, Gradio.
- **Data Engineering:** Python (Numpy,Pandas, PyTorch, TensorFlow, Keras), ETL Pipelines,Data Lake,Data Warehouse, SQL, Apache Spark, Databricks, OpenCV, Web Scraping,
- **Data Visualization:** Power Bi, Tableau

EXPERIENCE

Elevvo

Cairo, Egypt (Remote)

Machine Learning Intern

Sep 2025-Oct 2025

- Engineered and optimized K-Means clustering pipelines in Python, achieving a Silhouette Score of 0.82, and evaluated DBSCAN to assess model robustness and algorithm sensitivity for scalable customer segmentation.
- Deployed model outputs into an interactive Tableau interface, translating clustering results into production-ready business segments and reducing reporting overhead by 30%.

ITSOLERA PVT LTD

Islamabad, Pakistan (Remote)

AI & Machine Learning Intern

Jun 2025-Aug 2025

- Achieved 97% accuracy in pixel-level lane segmentation by architecting a VGG16-based U-Net model in TensorFlow/Keras; reduced inference latency by 15% compared to traditional CV (Canny/Hough) while maintaining high reliability in low-light conditions

PROJECTS

Predictive Maintenance of a Conveyor Belts Using Digital Twin Technology

Jamshoro, Pakistan)

Team Lead

Dec 2024-Nov 2025

- Reduced potential machine downtime by building a real-time fault forecasting system (Random Forest, 92.8% accuracy) using high-frequency IoT data (vibration, temperature, load) to predict failures early.
- Built a Digital Twin in Unity with Flask API, syncing real-time IoT data; set up PostgreSQL for historical telemetry and an interactive live dashboard.
- Mitigated hardware risks by implementing automated safety logic to trigger emergency motor stops and dashboard freezes when faults (≥ 500) were breached, preventing critical failure.

NeuroBot-AI: Smart Learning Assistant

Project Lead

- Built a RAG-based chatbot using Meta-Llama-3 and FAISS, achieving $<2.3s$ response time for deep learning queries, while acting as an AI mentor with learning path recommendations and ensuring fully verifiable outputs using page-level attribution and MLflow.

Telcomonicuation Customer Churn Prediction System

MLOps Developer

- Minimized customer attrition risk by developing a production-ready XGBoost churn model achieving 92.51% Recall, integrated with MLflow for experiment tracking and FastAPI/Gradio for real-time inference.
- Engineered a CI/CD pipeline using GitHub Actions and Docker to automate containerized image builds, ensuring consistent delivery of the model to production.

EDUCATION

Mehran University of Engineering and Technology

Jamshoro, Pakistan

Bachelor of Engineering in Computer Systems Engineering

Graduated November 2025

ACTIVITIES & LEADERSHIP

- IEEE Student Chapter: Lead for BCI-based project "The NeuroVoice" in EPICS IEEE Competition.
- Microsoft Learn Student Ambassador: Mentored 40+ participants on Exploratory Data Analysis (EDA) using Python.